Claim Processing Pipeline Assignment

Goal

Build a FastAPI service that processes PDF claims using LangGraph to orchestrate document segregation and multi-agent extraction.

Requirements

API Endpoint

POST `/api/process`

Input:

- `claim_id` (string)
- `file` (PDF)

Output: Any JSON with extracted data

LangGraph Workflow

Build this exact flow:

START → [Segregator Agent (AI)] → [ID Agent] → [Aggregator] → END
↓ → [Discharge Summary Agent] ↗
(classifies pages → [Itemized Bill Agent] ↗
into 9 doc types,
routes to agents)

Nodes:

1. **Segregator Agent (AI-powered)**
    - Takes PDF file
    - Uses LLM to analyze and classify pages into document types:

        - claim_forms
        - cheque_or_bank_details
        - identity_document
        - itemized_bill
        - discharge_summary
        - prescription
        - investigation_report
        - cash_receipt
        - other
    - Routes pages to appropriate extraction agents
2. **Extraction Agent Nodes** (3 required)

   Route the necessary pages to separate agents without passing the whole pdf

    - **ID Agent** - Extracts identity information (patient name, DOB, ID numbers, policy details)
    - **Discharge Summary Agent** - Extracts diagnosis, admit/discharge dates, physician details
    - **Itemized Bill Agent** - Extracts all items with costs and calculates total amount
    - Each agent processes only the pages assigned to them by segregator
3. **Aggregator Node**
    - Combines all agent results
    - Returns final JSON with all extracted data

Key Rule: Segregator classifies pages into document types. Only 3 extraction agents process the relevant pages.

Submission

Provide:

1. **Live API URL (Optional)** - Deploy if you want (Render/Railway/Fly.io)
2. **Video Explanation (Preffered)** - Loom or any video platform explaining:
    - Your LangGraph workflow
    - How the segregator agent works
    - How extraction agents process their assigned pages
    - The complete process flow
3. **GitHub Repo** with README

Tech Stack

- FastAPI
- LangGraph